

Azure DevOps

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Senior Cloud Architect

Agenda



Overview Azure DevOps
Demo Workflow
Automate Deployments

Wolfgang Ofner

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Focus on Azure, Kubernetes, and DevOps



[wolfgangofner](https://www.linkedin.com/in/wolfgangofner)



YouTube

[@programmingwithwolfgang](https://www.youtube.com/@programmingwithwolfgang)



Organization Settings

ProgrammingWithWolfgang

+ New project

ProjectsMy work itemsMy pull requests

Filter projects

 MicroserviceDemo

 Presentation

 Youtube

 AzureArcDemo

 AzureArcDemo	
 K8sAdoAgent	
 MicroserviceDemo	
 Presentation	
 WindowsDeveloperArcDemo	
 Youtube	

Organization Settings

Manage global settings:

- Manage users and their licenses
- Configure billing
- Extensions
- Microsoft Entra
- Processes
- Global settings for repositories

Access Level

Basic

- Provides access to all features except Test Plans

Access Level

Basic

- Provides access to all features except Test Plans

Visual Studio Subscriber

- Same features as Basic
- "Free" license
- Visual Studio Enterprise allows for an additional parallel build

Access Level

Basic + Test Plans

- Same as Basic plus Test Plans

Access Level

Basic + Test Plans

- Same as Basic plus Test Plans

Stakeholder

- Unlimited free users
- Limited features including no repository or pipeline access

Billing

Microsoft hosted agents have 1800 minutes / month includes for free

Microsoft hosted agents allow 1 parallel job for free

5 Basic licenses are included for free

Billing

Billing needs an Azure subscription

- More than 5 Basic licenses
- Additional build minutes
- Additional parallel jobs

Extensions

Free and paid extensions

Bring additional features to ADO

- Time tracking
- Pipeline tasks
- Integration with other tools, e.g. Jira

Microsoft Entra

Microsoft Entra, formerly Azure Active Directory
Used for user and billing management

Process

Manage existing project processes

- Scrum
- Agile
- CMMI
- Basic

Process

“Create inherited process” based on an existing process

- Change existing items and workflows
- Create new items and workflows

Agent Pools

Collection of (pipeline) agents

- Manage groups of agents instead of individual ones
- Group agents based on capabilities → operating system, installed software
- Overview of all executed jobs

Agent Pools

Microsoft offers one agent in the Azure Pipelines pool for free

Microsoft-hosted agents vs. Self-hosted agents

Parallel Jobs

Overview of available jobs

Purchase additional jobs

- Microsoft-hosted agent: 40 USD / month
- Self-hosted agent: 15 USD / month

Project Settings

Permissions

Configure permissions for users or groups
(groups are preferred)

- Allow
- Deny
- Not set (has a deny effect)

Permissions

Permissions can be inherited

Direct permissions win over inherited ones

Service Hooks

Integrate third party tools with ADO

- Teams
- Slack
- Azure Service Bus
- Datadog
- Grafana

Service Hooks

Supported events

- Work item update
- Code pushed
- Build complete
- Release deployed

Project Configuration

Configure settings for the project

- Create and configure iterations
- Create and manage areas of the project

Team Configuration

Configure settings for individual teams

- Team settings such as backlog, workdays and bug handling
- Manage iterations
- Manage areas
- Create templates for work items

Service Connections

Configure connection to third party tools

- Azure
- DockerHub
- GitHub
- Jira
- NuGet
- Npm

Project Menu Items

Overview

Summary

- Provides project description
- Readme file or Wiki

Overview

Dashboard

- Create dashboards
- Pipeline status, work item count, burndown chart, etc.

Overview

Wiki

- Document where the code resides
- Link work items and code commits
- Uses Git in the background
- Markdown

Boards

Manage work items with different boards

- Boards
- Backlog
- Sprints
- Queries
- Delivery Plans

Boards

Plan team capacity per sprint

Burndown chart for sprints

Boards

Create custom queries

- Work items where status change > 180 days
- All closed work items in a timeframe
- Work items approved by team member XYZ

Boards

Delivery plans

- Visualize milestones and delivery plan over all teams
- Update iterations
- Used by Product Owner / Release Manager

Azure DevOps Features and Stories

Azure DevOps team all up



Settings



End Q2 21

today

Publish

> Teams	< September 2020	October 2020	November 2020	December 2020	January	February	>
> Azure DevOps Features	2020 - 9/30 + 3 Feature	10_2020 10/1 - 10/30 + 🏆 5 Feature	11_2020 11/1 - 11/30 + 🏆 15 Feature	12_2020 12/1 - 12/31 + 🏆 31 Feature	01_2021 1/1 - 1/29 + 🏆 9 Feature	02_2021 2/1 - 2/26 + 🏆 4 Feature	>
> Azure Pipelines Stories	2020 - 9/30 + 28 User Story	10_2020 10/1 - 10/30 + 📄 18 User Story	11_2020 11/1 - 11/30 + 📄 17 User Story 🐛 1 Bug	12_2020 12/1 - 12/31 + 📄 22 User Story 🐛 1 Bug	01_2021 1/1 - 1/29 + 📄 30 User Story 🐛 1 Bug	02_2021 2/1 - 2/26 + 📄 1 User Story	>
> Azure Release Stories	2020 - 9/30 + 0 User Story	10_2020 10/1 - 10/30 + 0 User Story	11_2020 11/1 - 11/30 + 0 User Story	12_2020 12/1 - 12/31 + 📄 1 User Story	01_2021 1/1 - 1/29 + 📄 1 User Story	02_2021 2/1 - 2/26 + 0 User Story	>
> Azure Artifacts Stories	2020 - 9/30 + 8 User Story	10_2020 10/1 - 10/30 + 📄 3 User Story 🐛 1 Bug	11_2020 11/1 - 11/30 + 📄 9 User Story	12_2020 12/1 - 12/31 + 📄 7 User Story	01_2021 1/1 - 1/29 + 📄 5 User Story	02_2021 2/1 - 2/26 + 0 User Story	>

Repos

Overview of everything Git related

- Files
- Commits
- Branches
- Tags
- Pull Requests

Repos

Advanced Security (GitHub Advanced Security for Azure DevOps)

- Comprehensive security suite
- Secret scanning
- Dependency scanning
- Static code scanning
- 49\$ per month per active committer → too expensive

Pipelines

Pipelines

- YAML pipelines
- Used for build and deployments tasks
- File is stored in Git repo

Pipelines

Environments

- Can be created in a pipeline or portal
- Overview of when and what was deployed to a specific environment
- Configure approvals and checks before deployment is executed

Pipelines

Outdated features

- Releases
- Task groups
- Deployment groups

Pipelines

Library

- Create variable groups which can be shared across multiple pipelines
- Securely store sensitive files such as certificates
- However, using Azure Key Vault is the recommended way

Test Plans

Needs additional license

Used for manual testing and exploratory testing

- Test cases can be linked to work items
- Specify test configuration, e.g. operating system, browser, etc.
- Tester can add screenshots or screen recordings

Artifacts

Create private feeds for packages

- NuGet
- Npm
- Maven
- Pip

Git

Git

Distributed version control

Commit changes with a message

Creates a history of all commits

Git

Commits are made to a branch

Branches are individual work areas

Use .gitignore file to exclude files or extensions

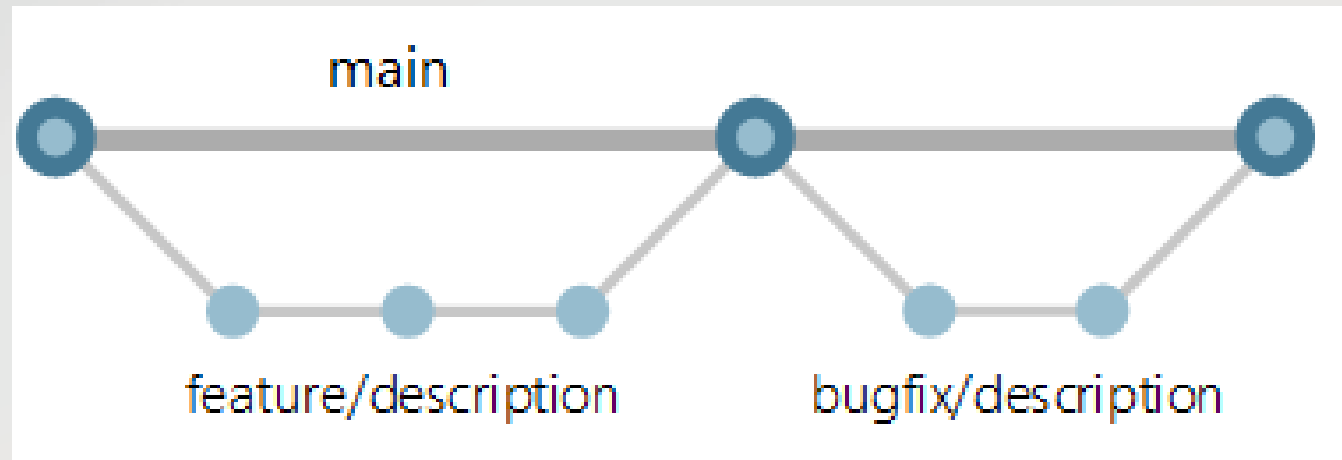
Git Branches

Main branch should only contain tested / deployable code

Developers work with "feature" branches

Once a feature is completed, feature branch will be merged into main branch

Git Branches

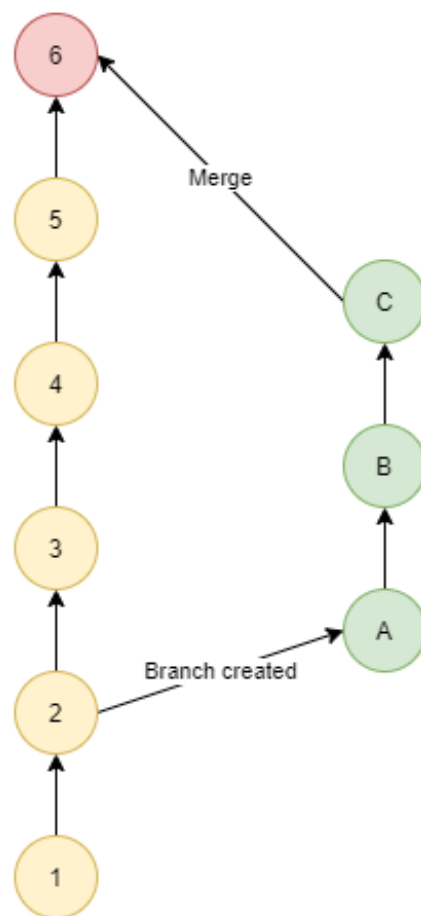


Git Branches

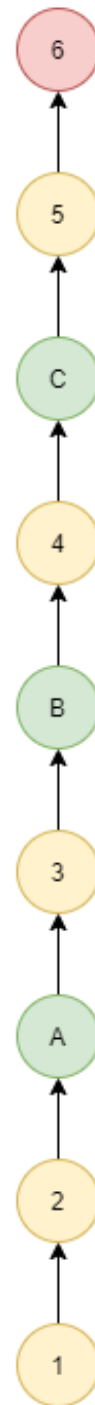
Various merge strategies are available

- Basic Merge

"Merge pull request #1 from branch"



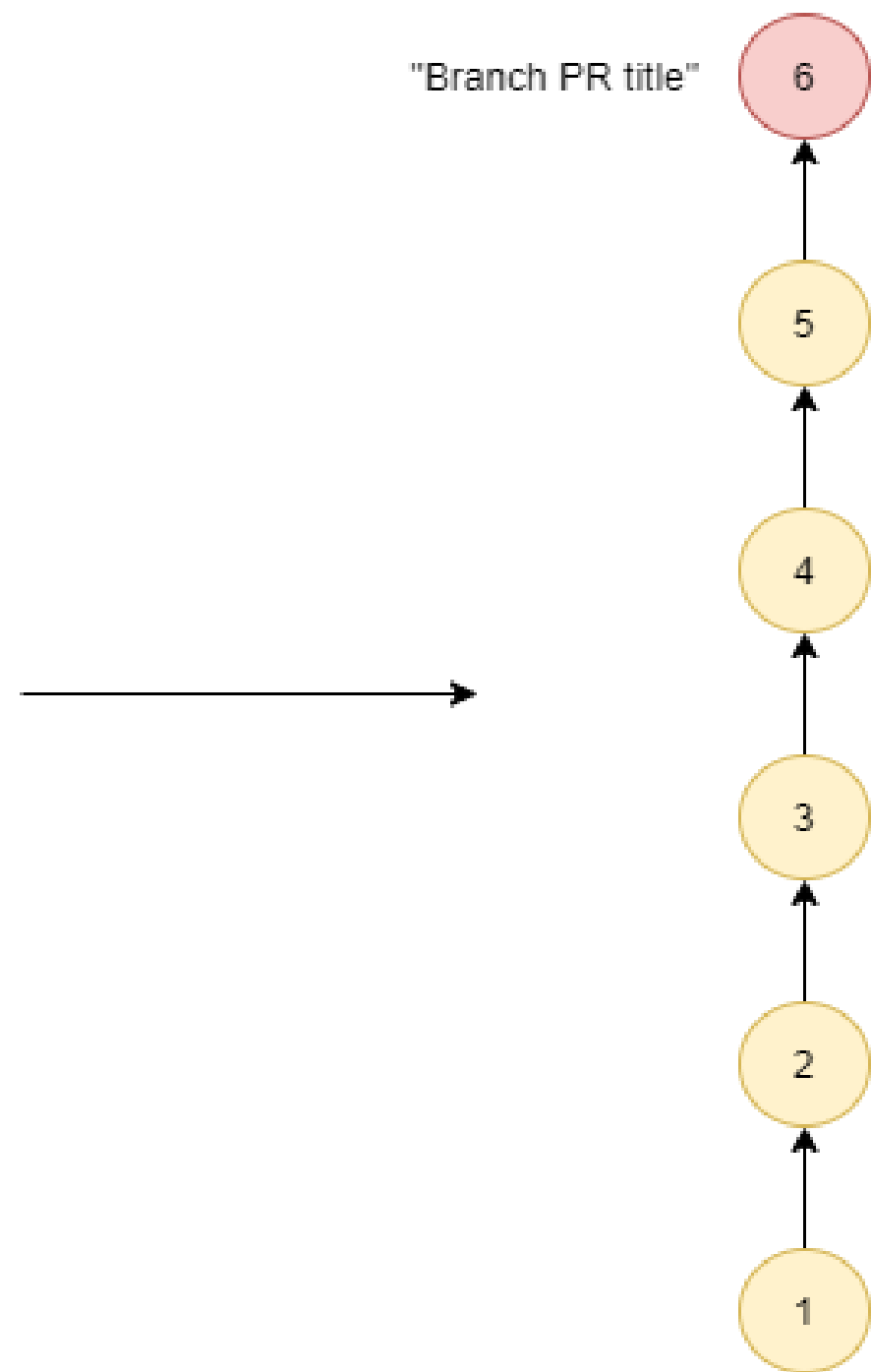
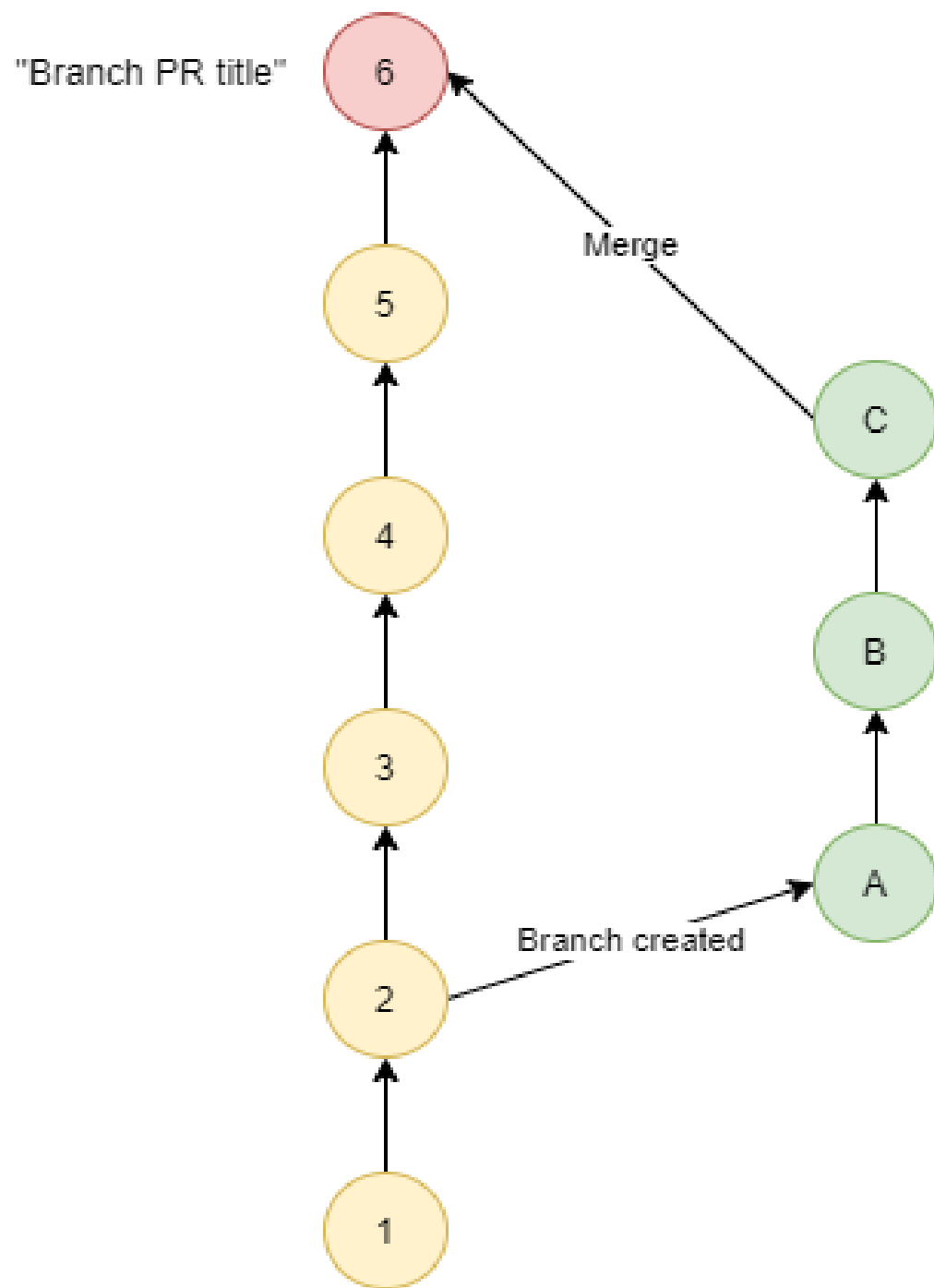
"Merge pull request #1 from branch"



Git Branches

Various merge strategies are available

- Basic Merge
- Squash Merge



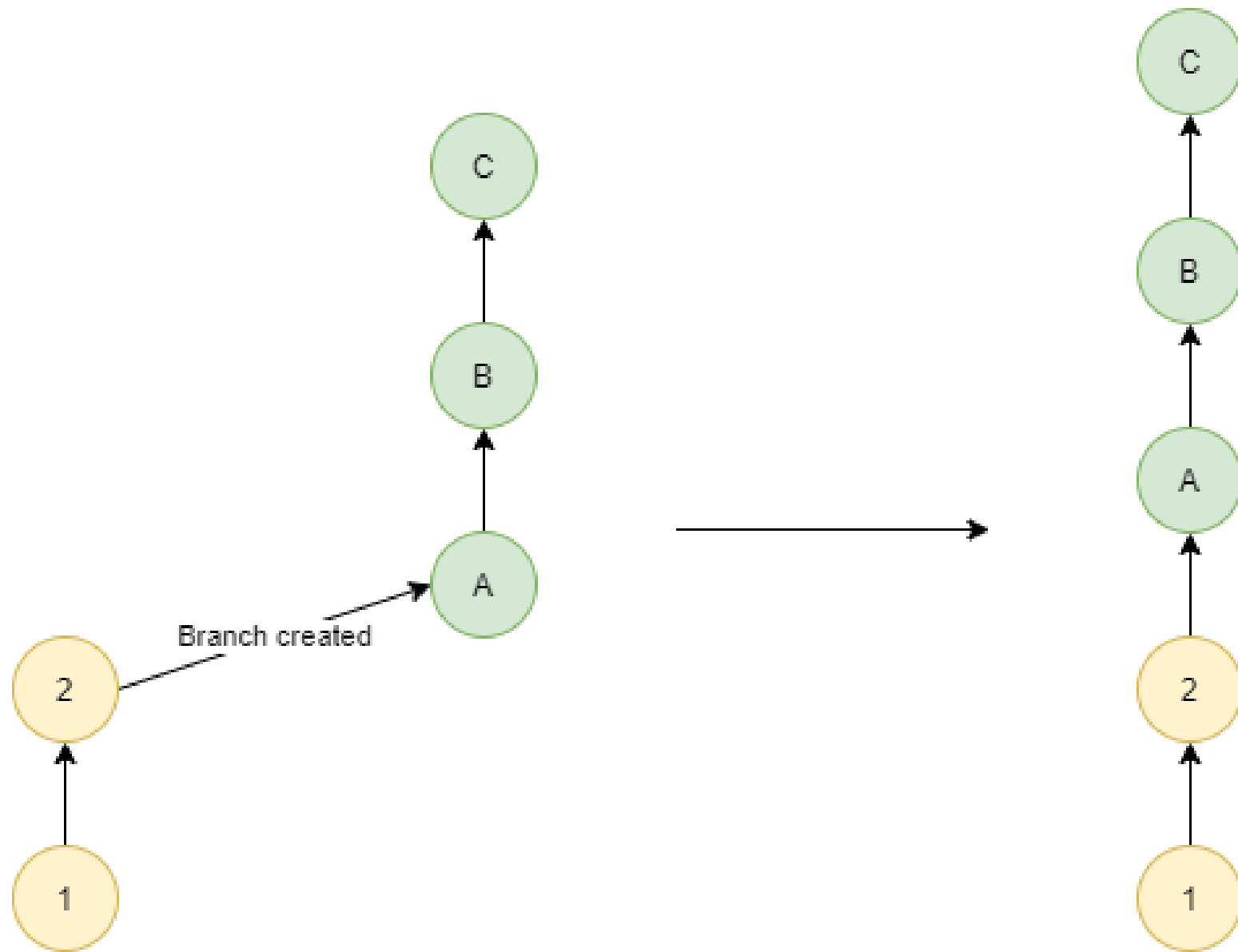
Git Branches

Various merge strategies are available

- Basic Merge
- Squash Merge
- Fast-Forward Merge

Base Branch

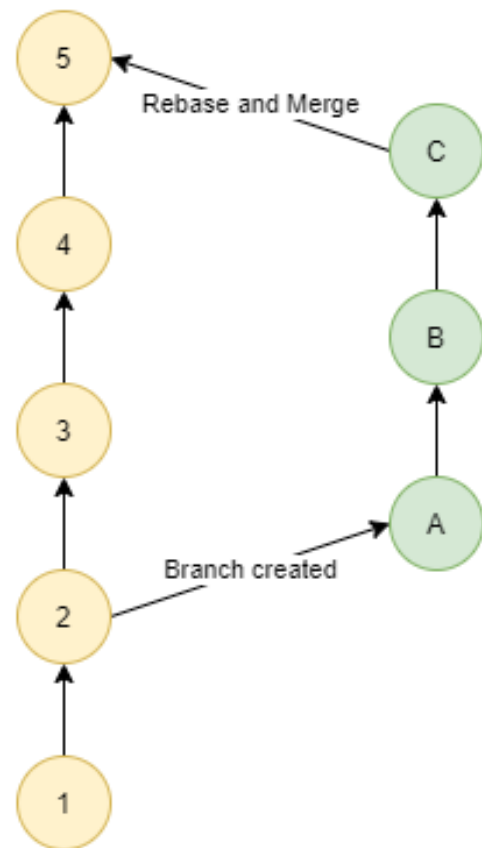
Branch Being Merged



Git Branches

Various merge strategies are available

- Basic Merge
- Squash Merge
- Fast-Forward Merge
- Rebase



Merge Conflicts

Git tries to determine the correct version if multiple developers change the same file

If Git can't figure it out → merge conflict and developers have to manually resolve it

Merge Conflicts

Pull destination branch and merge it locally to resolve the conflict

Use short-lived branches and merge often

Demo Project

P Presentation +

Overview

Boards

Repos

Files

Commits

Pushes

Branches

Tags

Pull requests

Advanced Security

Pipelines

Test Plans

Artifacts

Presentation is empty. Add some code!

Clone to your computer

HTTPS

SSH

`https://ProgrammingWithWolfgang@dev.azure.com/ProgrammingW`



OR



Clone in VS Code



Generate Git Credentials

Having problems authenticating in Git? Be sure to get the latest version [Git for Windows](#) or our plugins for [IntelliJ](#), [Eclipse](#), [Android Studio](#) or [Windows command line](#).

Push an existing repository from command line

HTTPS

SSH

`git remote add origin`

`https://ProgrammingWithWolfgang@dev.azure.com/ProgrammingWithWolfgang/Pre`



Import a repository

Import

Initialize master branch with a README or gitignore

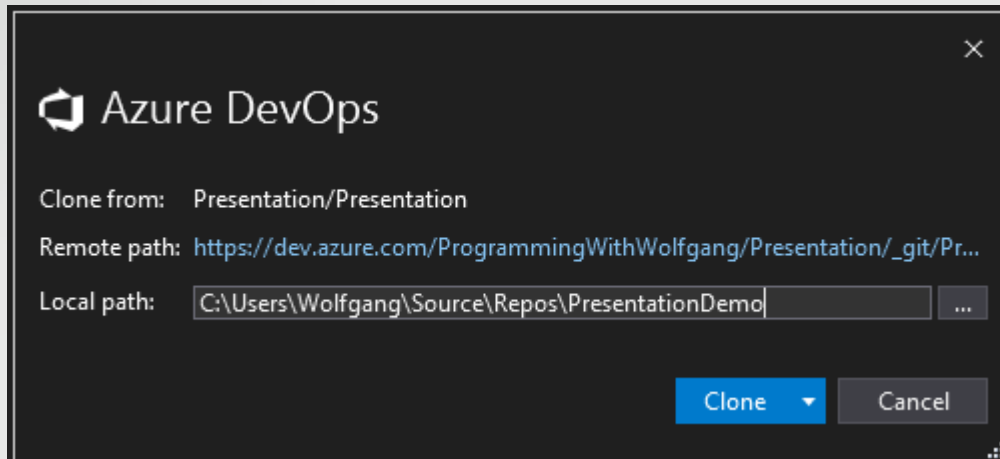


Add a README

Add a .gitignore: None



Initialize



Create a new project

Recent project templates

-  .NET Aspire Starter Application C# 
-  ASP.NET Core Web App (Model-View-Controller) C#
-  ASP.NET Core Web API C#
-  ASP.NET Core Web App (Razor Pages) C#

Search for templates (Alt+S)



All languages

All platforms

All project types



.NET Aspire Application

New

A project template for creating an empty .NET Aspire app.

C#

.NET Aspire

API

Cloud

Common

Service

Web

Web API



.NET Aspire App Host

New

A project template for creating a .NET Aspire app host (orchestrator) project.

C#

.NET Aspire

Cloud

Common



.NET Aspire Service Defaults

New

A project template for creating a .NET Aspire service defaults project.

C#

.NET Aspire

API

Cloud

Common

Service

Web

Web API



.NET Aspire Test Project (xUnit)

New

A project that contains xUnit.net integration tests of a .NET Aspire AppHost project.

C#

.NET Aspire

API

Cloud

Common

Service

Test

Web

Web API



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C#

Linux

macOS

Windows

Console

Next

Configure your new project

ASP.NET Core Web App (Model-View-Controller)

C#

Linux

macOS

Windows

Cloud

Service

Web

Project name

PresentationDemo

Location

C:\Users\Wolfgang\source\repos\PresentationDemo\

...

Solution

Create new solution

Solution name 

PresentationDemo

☐

Place solution and project in the same directory

Project will be created in "C:\Users\Wolfgang\source\repos\PresentationDemo\PresentationDemo\PresentationDemo\"

Back

Next

Additional information

ASP.NET Core Web App (Model-View-Controller)

[C#](#)[Linux](#)[macOS](#)[Windows](#)[Cloud](#)[Service](#)[Web](#)

Framework 

.NET 8.0 (Long Term Support) 

Authentication type 

None 

☐ Configure for HTTPS 

☐ Enable container support 

Container OS 

Linux 

Container build type 

Dockerfile 

☐ Do not use top-level statements 

☐ Enlist in .NET Aspire orchestration 

[Back](#)[Create](#)

.gitignore

```
1  ## Ignore Visual Studio temporary files, build results, and
2  ## files generated by popular Visual Studio add-ons.
3  ##
4  ## Get latest from https://github.com/github/gitignore/blob/master/VisualStudio.gitignore
5
6  # User-specific files
7  *.rsuser
8  *.suo
9  *.user
10 *.userosscache
11 *.sln.docstates
12
13 # User-specific files (MonoDevelop/Xamarin Studio)
14 *.userprefs
15
16 # Build results
17 [Dd]ebug/
18 [Dd]ebugPublic/
19 [Rr]elease/
20 [Rr]eleases/
21 x64/
22 x86/
23 [Aa][Rr][Mm]/
24 [Aa][Rr][Mm]64/
25 bld/
26 [Bb]in/
27 [Oo]bj/
28 [Ll]og/
29
30 # Visual Studio 2015/2017 cache/options directory
31 .vs/
```

Git Changes - PresentationDemo

master

↑↓ 0 / 0 [View all commits](#)

added first project and .gitignore

Commit All

☐ Amend

▸ Related Work Items

▴ Changes (89)

▴ C:\Users\Wolfgang\source\repos\Prese...

▴ PresentationDemo

▴ pipelines

▴ Examples

▴ templates

YML BuildVersioning.yml A

YML CreateHelmPackage.yml A

YML DatabaseDeploy.yml A

YML DeployHelmPackage.yml A

YML DockerBuildAndPush.y... A

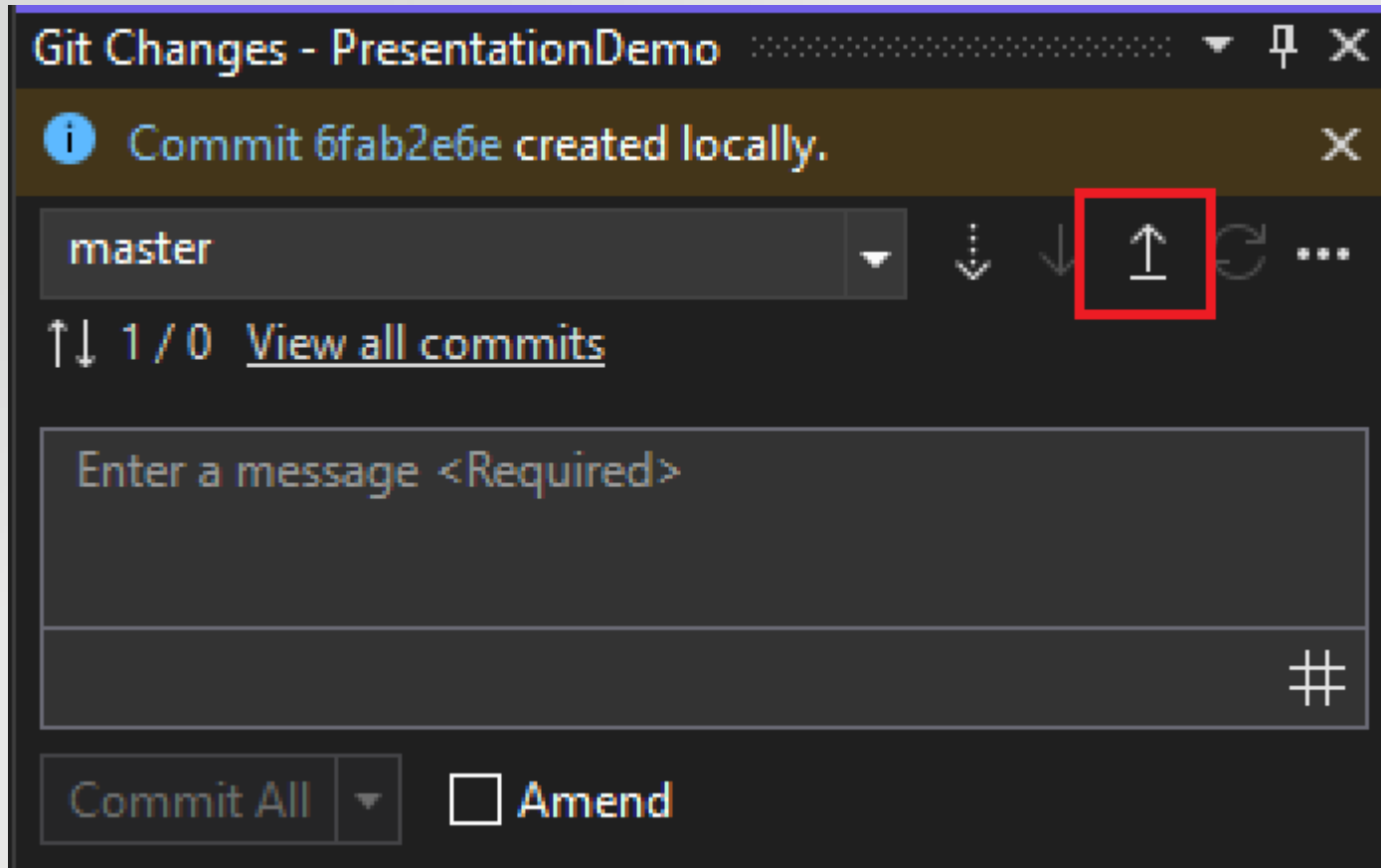
YML GetPrId.yml A

YML GetServiceBusConnecti... A

YML CustomerApi-CD.yml A

YML CustomerApi-Cl.yml A

YML NetCore-Cl-azure-pipeline.yml A



[Contents](#) [History](#)

Name ↑	Last change	Commits
 PresentationDemo	5m ago	6fab2e6e added first project and .gitignore Wolfga...
 .gitignore	5m ago	6fab2e6e added first project and .gitignore Wolfga...

Presentation +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts



Create your first Pipeline

Automate your build and release processes using our wizard, and go from code to cloud-hosted within minutes.

Create Pipeline :



Presentation



Overview



Boards



Repos



Pipelines



Pipelines



Environments



Releases



Library



Task groups



Deployment groups



Test Plans

Connect

Select

Configure

Review

New pipeline

Where is your code?



Azure Repos Git

YAML

Free private Git repositories, pull requests, and code search



Bitbucket Cloud

YAML

Hosted by Atlassian



GitHub

YAML

Home to the world's largest community of developers



GitHub Enterprise Server

YAML

The self-hosted version of GitHub Enterprise



Other Git

Any generic Git repository



Subversion

Centralized version control by Apache

[Use the classic editor](#) to create a pipeline without YAML.

 Presentation +

 Overview

 Boards

 Repos

 **Pipelines**

 Pipelines

 Environments

 Releases

 Library

 Task groups

 Deployment groups

✓ Connect

Select

Configure

Review

New pipeline

Select a repository

≡ Filter by keywords

Presentation ▼ ✕



Presentation

**Presentation**

Overview



Boards



Repos

**Pipelines**

Pipelines



Environments



Releases



Library



Task groups



Deployment groups



Test Plans



Artifacts

**Xamarin.iOS**

Build a Xamarin.iOS project.

**Starter pipeline**

Start with a minimal pipeline that you can customize to build and deploy your code.

**Existing Azure Pipelines YAML file**

Select an Azure Pipelines YAML file in any branch of the repository.

**.NET Core Function App to Windows on Azure**

Build a .NET Core function app and deploy it to Azure as a Windows function App.

**Android**

Build your Android project with Gradle.

**Ant**

Build your Java projects and run tests with Apache Ant.

**ASP.NET Core**

Build and test ASP.NET Core projects targeting .NET Core.

**C/C++ with GCC**

Build your C/C++ project with GCC using make.

**Deploy to Azure Kubernetes Service**

Build and push image to Azure Container Registry; Deploy to Azure Kubernetes Service

**Docker**

Build a Docker image

**Docker**

Build and push an image to Azure Container Registry

- P Presentation +
- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Releases
- Library
- Task groups
- Deployment groups
- Test Plans
- Artifacts

- ✓ Connect
- ✓ Select
- ✓ Configure
- Review

New pipeline

Review your pipeline YAML

Variables

Save and run

Presentation / azure-pipelines.yml *

Show assistant

```
1 # ASP.NET Core
2 # Build and test ASP.NET Core projects targeting .NET Core.
3 # Add steps that run tests, create a NuGet package, deploy, and more:
4 # https://docs.microsoft.com/azure/devops/pipelines/languages/dotnet-core
5
6 trigger:
7   - master
8
9 pool:
10   vmImage: ubuntu-latest
11
12 variables:
13   buildConfiguration: 'Release'
14
15 steps:
16   - script: dotnet build --configuration $(buildConfiguration)
17     displayName: 'dotnet build $(buildConfiguration)'
18
```

New pipeline

Configure your pipeline

-  **ASP.NET**
Build and test ASP.NET projects.
-  **ASP.NET Core (.NET Framework)**
Build and test ASP.NET Core projects targeting the full .NET Framework.
-  **.NET Desktop**
Build and run tests for .NET Desktop or Windows classic desktop solutions.
-  **Universal Windows Platform**
Build a Universal Windows Platform project using Visual Studio.
-  **Xamarin.Android**
Build a Xamarin.Android project.
-  **Xamarin.iOS**
Build a Xamarin.iOS project.
-  **Starter pipeline**
Start with a minimal pipeline that you can customize to build and deploy your code.
-  **Existing Azure Pipelines YAML file**
Select an Azure Pipelines YAML file in any branch of the repository.

Show more

Select an existing YAML file

Select an Azure Pipelines YAML file in any branch of the repository.

Branch
master

Path
/PresentationDemo/pipelines/NetCore-CI-azure-pipeline.yml

Select a file from the dropdown or type in the path to your file

[Presentation](#)

Cancel

Continue

✓ Connect

✓ Select

✓ Configure

Review

New pipeline

Review your pipeline YAML

Variables

Run



Presentation / PresentationDemo/pipelines/NetCore-CI-azure-pipeline.yml

Show assistant

```
1 |name: NetCore-PresentationDemo-CI
2 |trigger: none
3
4 |pool:
5 |  vmImage: 'ubuntu-latest'
6
7 |variables:
8 |  buildConfiguration: 'Release'
9
10 |steps:
  Settings
11 |  - task: DotNetCoreCLI@2
12 |    inputs:
13 |      command: 'restore'
14 |      projects: '**/PresentationDemo*.csproj'
15 |      displayName: 'Restore NuGet Packages'
16
  Settings
17 |  - task: DotNetCoreCLI@2
18 |    inputs:
19 |      command: 'build'
20 |      projects: '**/PresentationDemo*.csproj'
21 |      arguments: '--no-restore'
22 |      displayName: 'Build projects'
23
  Settings
24 |  - task: DotNetCoreCLI@2
25 |    inputs:
26 |      command: 'test'
27 |      projects: '**/*Test.csproj'
28 |      arguments: '--no-restore --no-build'
29 |      displayName: 'Run Tests'
30
  Settings
31 |  - task: DotNetCoreCLI@2
32 |    inputs:
33 |      command: 'publish'
34 |      publishWebProjects: false
35 |      projects: '**/PresentationDemo.csproj'
36 |      arguments: '--configuration $(buildConfiguration) --no-restore'
37 |      modifyOutputPath: false
38 |      displayName: 'Publish PresentationDemo'
39 |      condition: and(succeeded(), ne(variables['Build.Reason'], 'PullRequest'))
```

#NetCore-PresentationDemo-CI • added first project and .gitignore

Presentation

Cancel

Summary

Code Coverage

Manually run by  Wolfgang Ofner

View 7 changes

Repository and version

◆ Presentation

🔗 master 🔗 ac557633

Time started and elapsed

🕒 Just now

-

Related

📁 0 work items

📦 0 artifacts

Tests and coverage

🔗 Get started

Jobs		
Name	Status	Duration
 Job	Queued	

←

Jobs in run #NetCore-Presenta...
Presentation

Jobs	
▼  Job	25s
 Initialize job	1s
 Checkout Presentation@master to s	1s
 Restore Nuget Packages	10s
  Build projects	12s
 Run Tests	
 Publish PresentationDemo	
 Post-job: Checkout Presentation@m...	

 Build projects

🔍

⋮

```
1 Starting: Build projects
2 =====
3 Task      : .NET Core
4 Description : Build, test, package, or publish a dotnet application, or run a custom dotnet command
5 Version    : 2.242.0
6 Author     : Microsoft Corporation
7 Help       : https://docs.microsoft.com/azure/devops/pipelines/tasks/build/dotnet-core-cli
8 =====
9 Info: .NET Core SDK/runtime 2.2 and 3.0 are now End of Life(EOL) and have been removed from all hosted agents. If you're using these SDK/runtimes on hosted agents, kindly upgrade to newer versions which are not EOL, or else use UseDotNet task to install the required version.
10 /usr/bin/dotnet build /home/vsts/work/1/s/PresentationDemo/PresentationDemo.Test/PresentationDemo.Test.csproj -dl:CentralLogger,"/home/vsts/work/_tasks/DotNetCoreCLI_5541a522-603c-47ad-91fc-a4b1d163081b/2.242.0/dotnet-build-helpers/Microsoft.TeamFoundation.DistributedTask.MSBuild
11   PresentationDemo.Test -> /home/vsts/work/1/s/PresentationDemo/PresentationDemo.Test/bin/Debug/net8.0/PresentationDemo.Test.dll
12
13 Build succeeded.
14   0 Warning(s)
15   0 Error(s)
16
17 Time Elapsed 00:00:07.70
18 /usr/bin/dotnet build /home/vsts/work/1/s/PresentationDemo/PresentationDemo/PresentationDemo.csproj -dl:CentralLogger,"/home/vsts/work/_tasks/DotNetCoreCLI_5541a522-603c-47ad-91fc-a4b1d163081b/2.242.0/dotnet-build-helpers/Microsoft.TeamFoundation.DistributedTask.MSBuild.Logger.dl
```

← Jobs in run #NetCore-Presenta...
Presentation

Jobs

✓ Job37s

Initialize job1s

✓ Checkout Presentation@master to s1s

✓ Restore Nuget Packages10s

✓ Build projects14s

✓ Run Tests5s

✓ Publish PresentationDemo3s

✓ Post-job: Checkout Presentation...<1s

Finalize Job<1s

Report build status<1s

✓ Job

1 Pool: [Azure Pipelines](#)

2 Image: ubuntu-latest

3 Agent: Hosted Agent

4 Started: Just now

5 Duration: 37s

6

7 ▶ Job preparation parameters

42 [100% tests](#) passed

43 Job live console data:

44 **Finishing: Job**

ⓘ This run is being retained as one of 3 recent runs by master (Branch).

Summary Tests Code Coverage

Summary

1 Run(s) Completed (1 Passed, 0 Failed)

1

Total tests

+1



1 ● Passed

0 ● Failed

0 ● Others

100%

Pass percentage

↑ 100%

904ms

Run duration ⓘ

↑ +904ms

0

Tests not reported

🐛 Bug ▾

🔗 Link

☰ Filter by test or run name



Hooray! There are no test failures.

Change the test outcome filter to view tests relevant to you.

Pull Request Policies

Pull Request Policies

The main branch should always be in a deployable state

Create pull request policies to protect the main branch

Pull Request Policies

- Enforce reviewers
- Validate that the code builds before merging
- Run unit tests
- Enforce linking of work items (no work should be done without a work item)
- Deploy the application to a test environment
- Enforce merge strategies

Project Settings

MicroserviceDemo

General

Overview

Teams

Permissions

Notifications

Service hooks

Dashboards

Boards

Project configuration

Team configuration

GitHub connections

Pipelines

Agent pools

Parallel jobs

Settings

Test management

Release retention

Service connections

XAML build services

Repos

Repositories

Artifacts

Storage

Test

Retention

All Repositories

Filter by keywords

MicroserviceDemo

MicroserviceDemo

Settings

Policies

Security

Approvals and checks

Repository Policies

Off

Commit author email validation

Block pushes with a commit author email that does not match the following patterns.

Off

File path validation

Block pushes from introducing file paths that match the following patterns.

Off

Case enforcement

Avoid case-sensitivity conflicts by blocking pushes that change name casing on files, folders, branches, and tags. [Learn more](#)

Off

Reserved names

Block pushes that introduce files, folders, or branch names that include platform reserved names or incompatible characters. [Learn more](#)

Off

Maximum path length

Block pushes that introduce paths that exceed the specified length. [Learn more](#)

Off

Maximum file size

Block pushes that contain new or updated files larger than this limit.

Branch Policies

Protect important branch namespaces in this repository with pre-merge checks and policies

master

Default

Compare

Filter by keywords

master 🔍 Default Compare

Branch Policies

Note: If any required policy is enabled, this branch cannot be deleted and changes must be made via pull request.

☒ On

Require a minimum number of reviewers

Require approval from a specified number of reviewers on pull requests.

Minimum number of reviewers

1

- ☐ Allow requestors to approve their own changes
- ☐ Prohibit the most recent pusher from approving their own changes
- ☐ Allow completion even if some reviewers vote to wait or reject
- ☒ When new changes are pushed:
 - ☒ Require at least one approval on every iteration. [Learn more](#)
 - ☐ Require at least one approval on the last iteration
 - ☐ Reset all approval votes (does not reset votes to reject or wait)
 - ☐ Reset all code reviewer votes

☒ On

Check for linked work items

Encourage traceability by checking for linked work items on pull requests.

- ☒ Required
 - Block pull requests from being completed unless they have at least one linked work item.
- ☐ Optional
 - Warn if there are no linked work items, but allow pull requests to be completed.

☒ On

Check for comment resolution

Check to see that all comments have been resolved on pull requests.

- ☒ Required
 - Block pull requests from being completed while any comments are active.
- ☐ Optional
 - Warn if any comments are active, but allow pull requests to be completed.



On

Limit merge types

Control branch history by limiting the available types of merge when pull requests are completed.

Allowed merge types:



Basic merge (no fast-forward)

Preserves history exactly as it happened during development



Squash merge

Creates a linear history by condensing the source branch commits into a single new commit on the target branch



Rebase and fast-forward

Creates a linear history by replaying the source branch commits onto the target without a merge commit



Rebase with merge commit

Creates a semi-linear history by replaying the source branch commits onto the target and then creating a merge commit

Build Validation 1

Validate code by pre-merging and building pull request changes.

Enabled

Name ↑

Path filter

Trigger



On

CustomerApi-CI

Required

/CustomerApi/

Automatic

Expires after 12 hours

Enforce Pull Requests

Commit

×

⊗

TF402455: Pushes to this branch are not permitted; you must use a pull request to update this branch.

Comment

Updated JokeController.cs

Branch name

master

Work items to link

Search work items by ID or title

▼

Wiki

Wiki

Create documentation using markdown

Have your documentation in the same place as your code and work items

Seamlessly link work items and code commits

Wiki

Some items are only available when the repository is checked out

- .attachments folder with all uploaded files
- .order file with order of files

3rd Party Repository Access

3rd Party Repository Access

3rd parties might need access to your repository

- GitOps to monitor branches and deploy changes
- Access a private NuGet feed
- Manage Agent Pools

3rd Party Repository Access

Allow access with Personal Access Token

- Token has an expire date
- Define permissions as needed
- Token value is only displayed after creation

Create a new personal access token

Name

MyToken

Organization

ProgrammingWithWolfgang

Expiration (UTC)

30 days

8/18/2024

Scopes

Authorize the scope of access associated with this token

Scopes ☐ Full access

☒ Custom defined

Work Items

Work items, queries, backlogs, plans, and metadata

☒ Read ☒ Read & write ☐ Read, write, & manage

Code

Source code, repositories, pull requests, and notifications

☐ Read ☐ Read & write ☐ Read, write, & manage ☐ Full ☒ Status

Build

Artifacts, definitions, requests, queue a build, and update build properties

☐ Read ☐ Read & execute

Release

Read, update, and delete releases, release pipelines, and stages

☒ Read ☐ Read, write, & execute ☐ Read, write, execute, & manage

Test Management

Read, create, and update test plans, cases, and results

☐ Read ☐ Read & write

Packaging

Create, read, update, and delete feeds and packages

☐ Read ☐ Read & write ☐ Read, write, & manage

[Show all scopes](#) (29 more)

Create

Cancel

Success!

You have successfully added a new personal access token. Copy the token now!

MyToken token

hvvwaurczmthqlwwreftgr



Warning - Make sure you copy the above token now. We don't store it and you will not be able to see it again.

3rd Party Repository Access

Use public SSH keys

- 3rd party has the same access as the user who added the SSH key
- Use PATs whenever possible

SSH Public Keys

Azure DevOps

User settings

Wolfgang Ofner

Account

- Profile
- Time and Locale
- Permissions

Preferences

- Notifications
- Theme
- Usage

Security

- Personal access tokens
- SSH public keys
- Authorizations

SSH Public Keys

Connect to your Git repos through SSH Public Keys when you can't use the recommended Git Credential Managers or Personal Access Tokens to securely connect to

Compare the server fingerprint when you connect via Git to ensure you've connected to Azure DevOps. The fingerprint should match one of the following:

Server MD5 Fingerprint 97:70:33:82:fd:29:3a:73:39:af:6a:07:ad:f8:80:49 (RSA)

Server SHA256 Fingerprint ohD8VZEXGWo6Ez8GSEJQ9WpafgLFsOfLOtGGQCQo6Og (RSA)



There are no SSH public keys added to your account on this organization.

Visit the [documentation](#) to learn how to generate your first SSH Key and select New Key to add it to your account.

Add New SSH Key

Name*

MyKey

Public Key Data*

```
ssh-rsa
AAAAB3NzaC1yc2EAAAABIwAAAQEAklIOUpkDHrfHY17SbrmTlp
NLTGK9Tjom/BWDSU
GPI+nafzIHDTYW7hdI4yZ5ew18JH4JW9jbhUFrviQzM7xlELEVf4h
9IFX5QVkbPppSwg0cda3
Pbv7kOdJ/MTyBIWXFCR+HAo3FXRitBqxiX1nKhXpHAZsMciLq8V
6RjsNAQwdsdMFvSIVK/7XA
t3FaoJoAsncM1Q9x5+3V0Ww68/elFmb1zuUFijQJKprX88XypND
vjYNby6vw/Pb0rwert/En
```


One Project
vs
Multiple Projects

Project Management

One project per product with several teams
Microsoft recommendations

- Use single project due to lowered administrative effort
- Use multiple project if you want custom processes or different administrative policies
- Private vs. public projects

Azure DevOps

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Senior Cloud Architect

